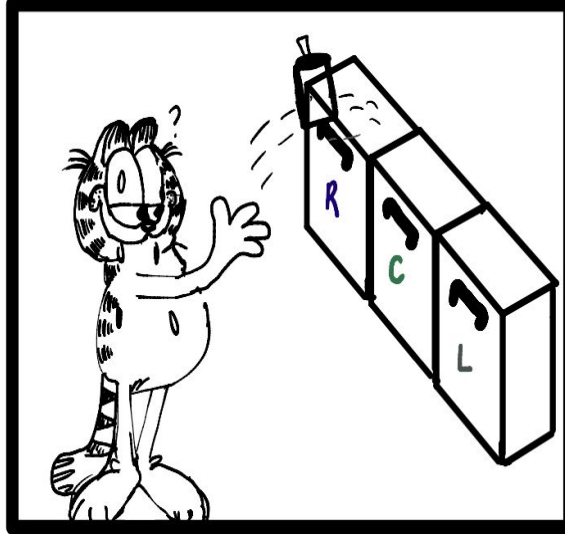
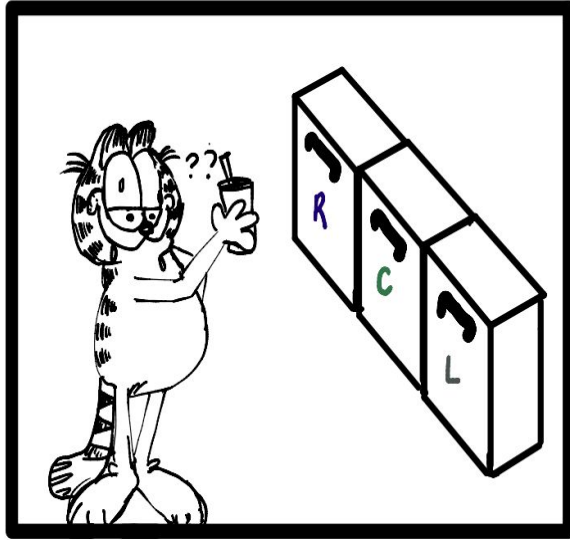


... EcoGate

Team 3: Rania, Rishi, Zainab, Elaine, Anya

Innovation Story and Learning Journey



Problem Statement

Despite having good intentions, busy college students frequently mis-sort their waste due to confusion, time pressure, and lack of feedback. This behavior contributes to contamination, wasted resources, and stalled progress toward campus sustainability goals.

Pain Points

Uncertainty → Landfill by Default

- When unsure or in a rush, students toss everything into the trash.
-
- “If I don’t know where it goes, I just throw it in the landfill bin.”

Fear of Doing It Wrong

Strict or unclear compost/recycle rules overwhelm users, causing them to give up.

“It’s too complicated—what if I mess it up?”

No Feedback → No Motivation

Without instant feedback or recognition, sorting feels thankless and invisible.

“There’s no way to know if I’m doing it right or if it matters.”

Key Insights

- Default to landfill when in doubt
- Fear of making mistakes
- No instant feedback or reward
 - Lack of recognition of effort
- Habits are hard to change
- Need for clear and in-context signs



Larger Consequences

- **Contamination:** Mixing recyclables or compostables with landfill trash leads to contamination, increasing pollution and wasting valuable resources.
- **Environmental Impact:** contamination complicates efforts to reduce greenhouse gas emissions, conserve natural resources, and protect ecosystems.
- **Long-Term Consequences:** Continued mismanagement of waste threatens environmental health and delays progress toward a sustainable future.

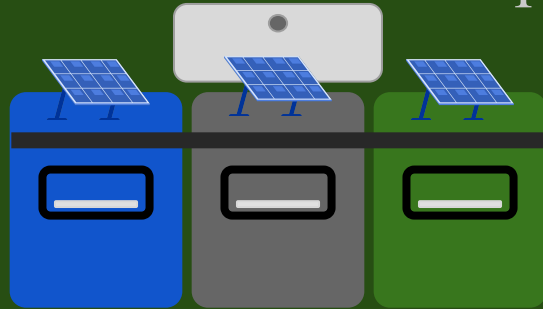


HMW Statement

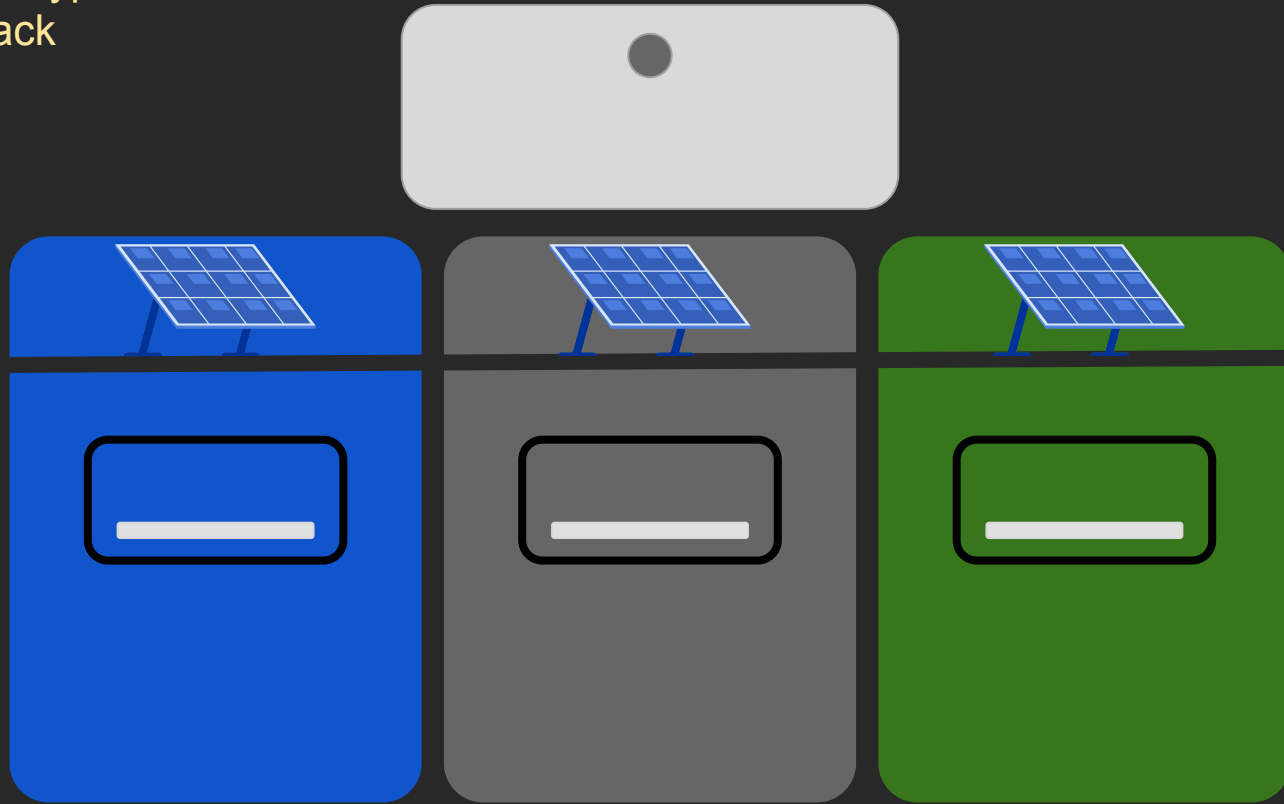
How might we integrate effortless education and encourage busy students to participate in proper waste disposal?

Solution

EcoGate scans students' waste and using AI technology, it can categorize it into the proper waste bin. Only after it analyzes waste, will the proper bin (landfill, recycle, or compost) open up for the student to dispose of their waste

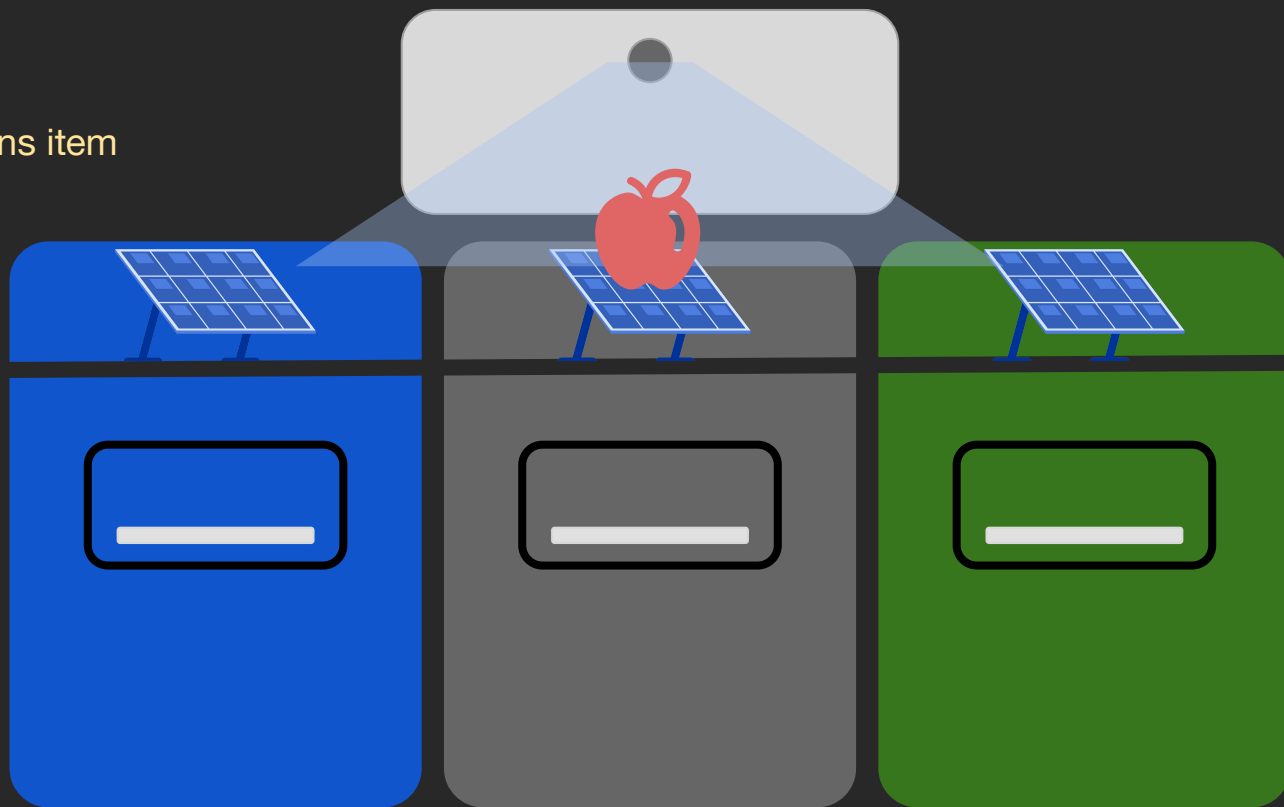


Updated prototype after
users feedback



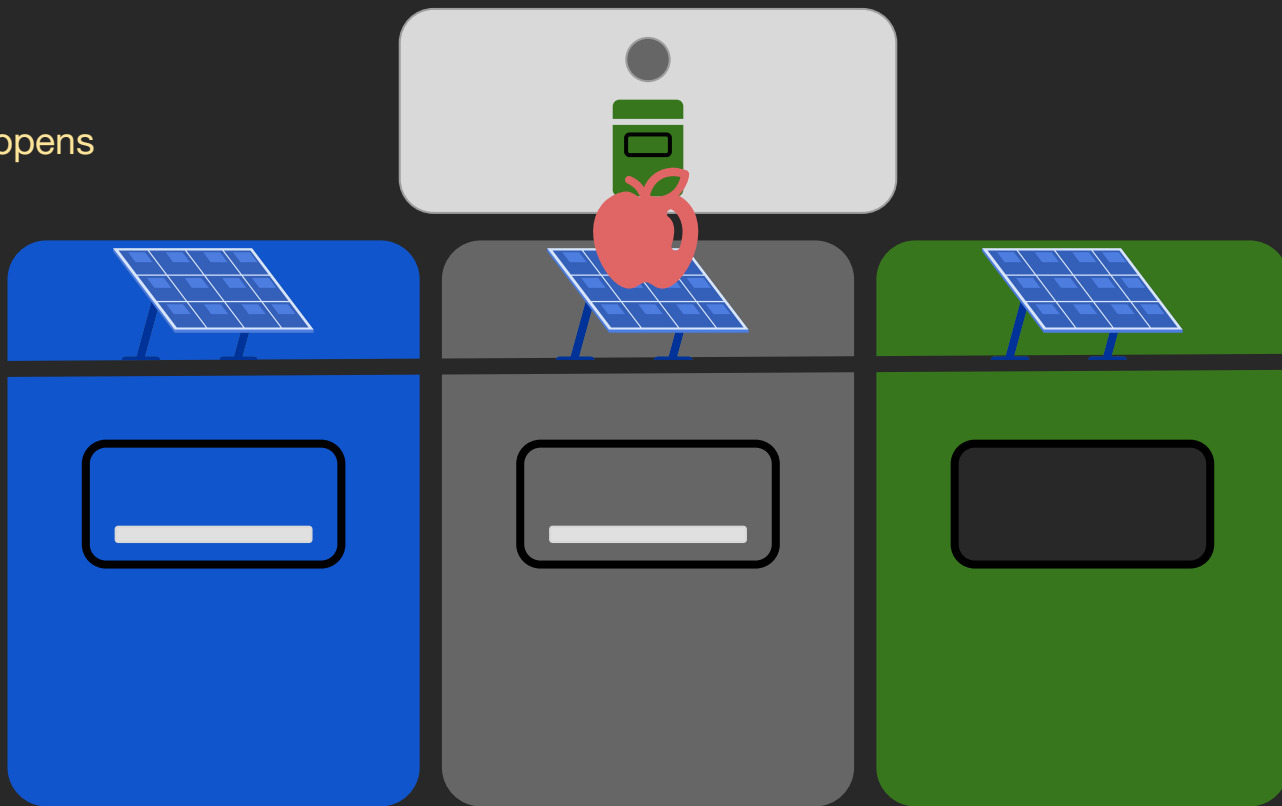
1

Camera scans item



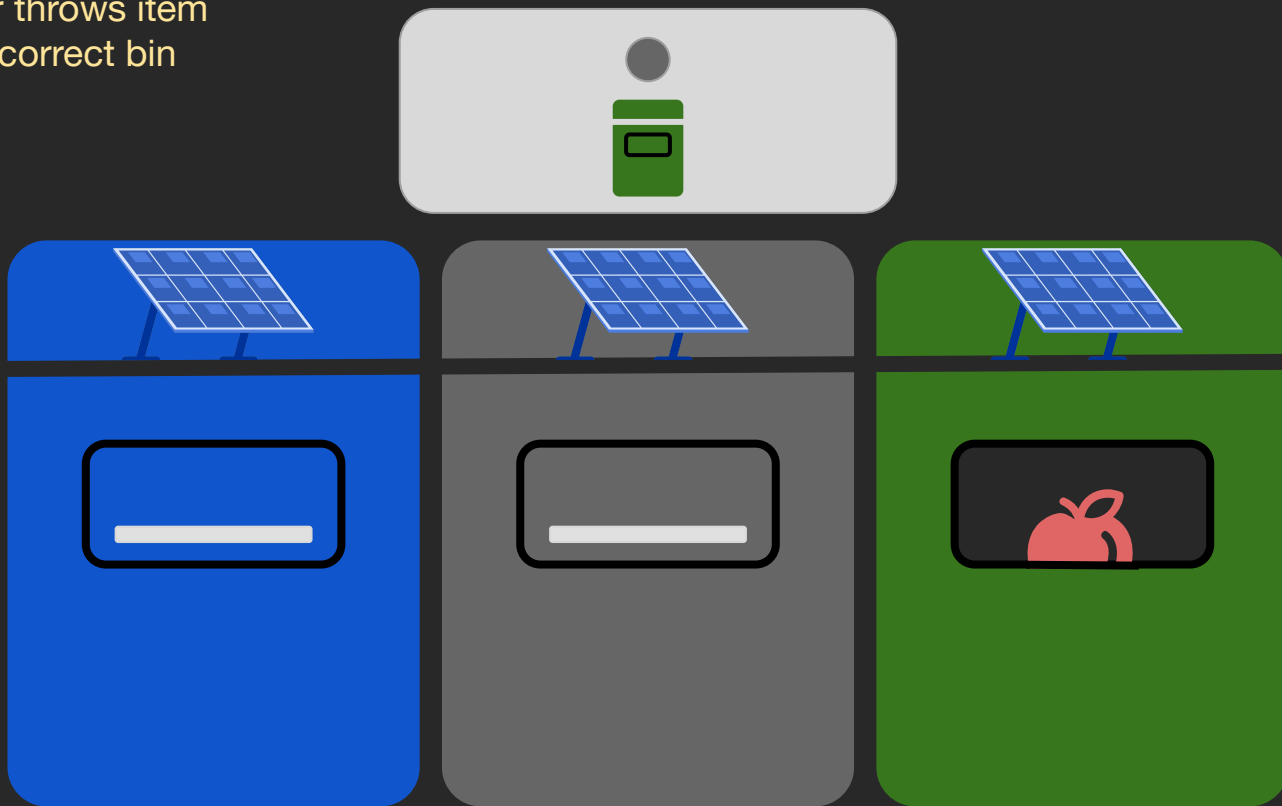
2

Correct bin opens



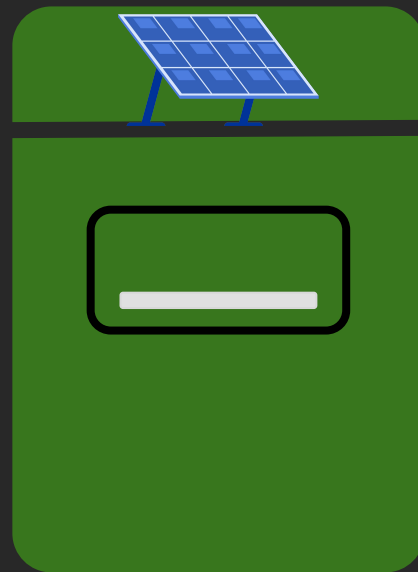
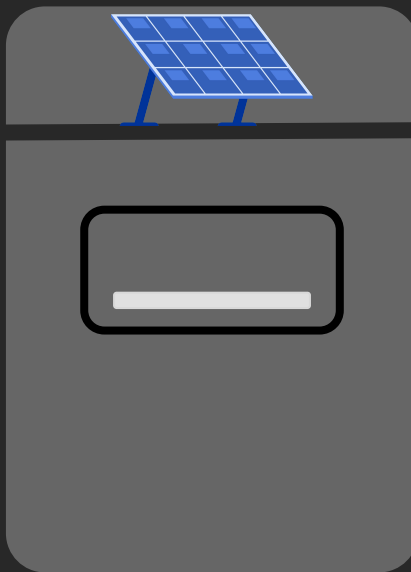
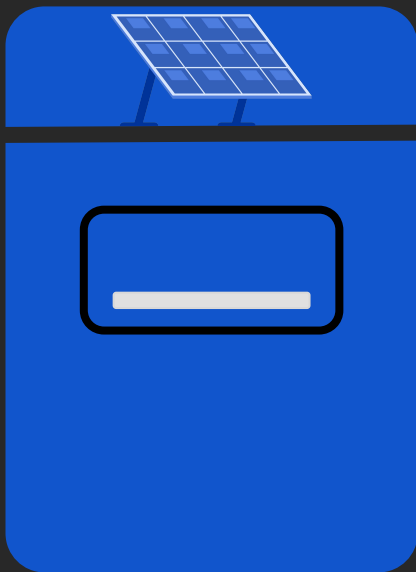
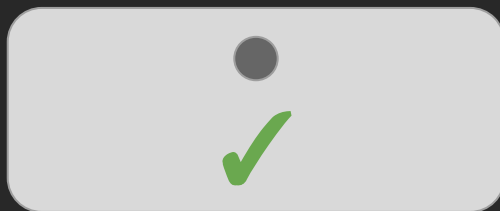
3

User throws item
into correct bin



4

Screen lights up & bin
closes



Prototype & User feedback

- Users highlighted its simplicity and “effort-free” sorting
- What happens when the bin is full ?
 - Sensor auto-alerts janitorial staff
- What happens if the mechanics malfunction?
 - Added manual-open handle

Business Model

- Value Proposition:
 - Waste sorting made simple for every user
 - Clear guidance on where each item belongs
 - Prevent bin contamination and reducing landfill waste

Business Model

- Customer Segments:
 - UC Berkeley students & staff (pilot)
 - Expand to other universities, schools & public venues
- Channels, Partners, & Relationships:
 - School campuses
 - UC Berkeley outreach
 - Local governments (city of Berkeley)
 - Smart-bin manufacturers
 - NGO & city/distributor partnerships
 - Green-tech & campus events
 - Impact stories on social media

Business Model

- Revenue & Costs:
 - Revenues: product sales, data monetization
 - Fixed Costs: designing new bins, developing & installing technologies on bins (scanner, sensors, solar panels)
 - Variable Costs: bin upkeep & maintenance, software updates

Thank You !

